Tabular Foundation Models for Data Science Agents

What Are Tabular Foundation Models?





DeepLearning.AI

What is tabular data? A concrete example

Which patients have early-stage Alzheimer (based on omics blood markers)?

ID	L-Carnitin	Creatinin	Homocystein	Beta-hydroxybutyrat	Early-stage Alzheimer
1	45.2	85	12.1	0.4	Yes
2	38.7	72	10.5	0.3	No
3	41.0	90	13.2	0.2	Yes
4	36.5	80	9.4	0.5	No
5	44.8	78	12.0	0.6	Yes
6	39.3	88	11.1	0.4	No
7	42.1	76	13.5	0.7	Yes
8	37.5	70	8.9	0.3	No
9	40.9	92	14.1	0.5	Yes
10	36.0	75	10.2	0.4	No
⋮	⋮	⋮	⋮	⋮	⋮
5000	43.0	84	12.7	0.6	Yes
5001	41.2	81	11.5	0.4	
5002	43.5	83	11.8	0.5	
5003	39.9	74	13.0	0.3	

X_{train}

y_{train}

X_{test}

y_{test}

Tabular prediction task

Further examples (among 1000s)

- Will this treatment work for this patient?
- How strong will this material be?
- What type of galaxy is this data from?
- Which product will this customer buy?
- Will this credit be paid back?
- ...

The decade-old problem formulation many of you saw in Machine Learning 101

Tabular foundation models solve a broad range of tasks

Example: treatment success prediction

Age	Existing Conditions (ICD)	Biomarker	Stage of Cancer	...	Responder ?
38	E11.9, I10	BRCA1+	IIIA	...	Yes
83	—	HER2+	IIB	...	Yes
71	N18.3, E78.5	EGFR-	IV	...	No
...	...	...	...	...	...
52	I10, E11.2	None	IIIA	...	?

Tabular foundation models solve a broad range of tasks

Example: credit risk assessment

Loan ID	FICO	Existing Products	...	Default ?
PR_PLM_PD_2401	750	3C	...	No
PR_PLM_PD_2501	530	2C1M	...	Yes
PD_PLA_PD_0459	820	1A	...	No
...	...	...	...	...
PD_PWL_PD_1021	650	2A2C	...	?

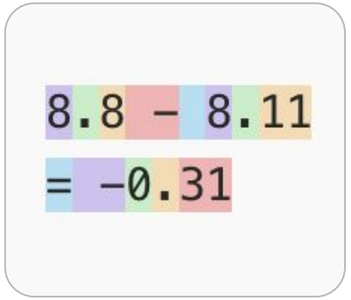
Tabular foundation models solve a broad range of tasks

Example: customer spend forecasting

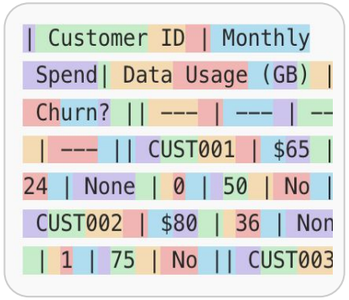
Lifetime value so far	Multiple accounts	Last Login Days	...	Spend Next Quarter
$1290.20	Yes	8	...	**$310.40**
$901.01	Yes	18	...	**$45.80**
$652.10	No	3	...	**$188.30**
...	...	...	...	...
$1147.49	No	8	...	**?**

LLMs aren't built for this

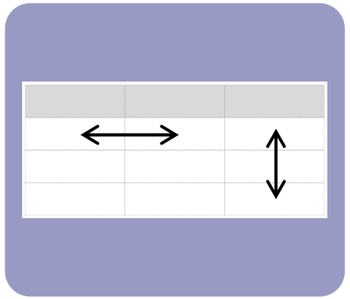
Why language models structurally fail on tabular data



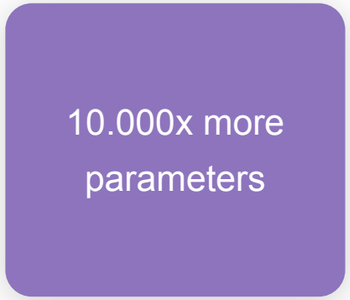
LLMs tokenize digits, don't understand entire numbers



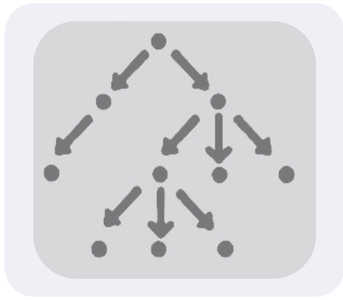
LLMs flatten tables into a (very long) sequence



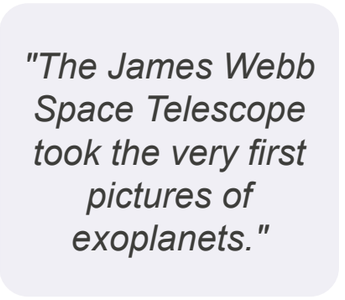
LLMs don't understand table invariances



Most LLM parameters are wasted on tables



LLMs are non-deterministic



LLMs hallucinate

LLMs are far too slow, expensive and inaccurate for tabular predictions

LLMs

Tasks over **unstructured** data, no matter the domain

OpenAI ANTHROPIC MISTRAL AI_ cohere

- Excel at *language & knowledge*
- Generate *summaries, explanations, code*
- **Boost business productivity**

Tabular Foundation Models (TFMs)

Tasks over **structured** data, no matter the domain

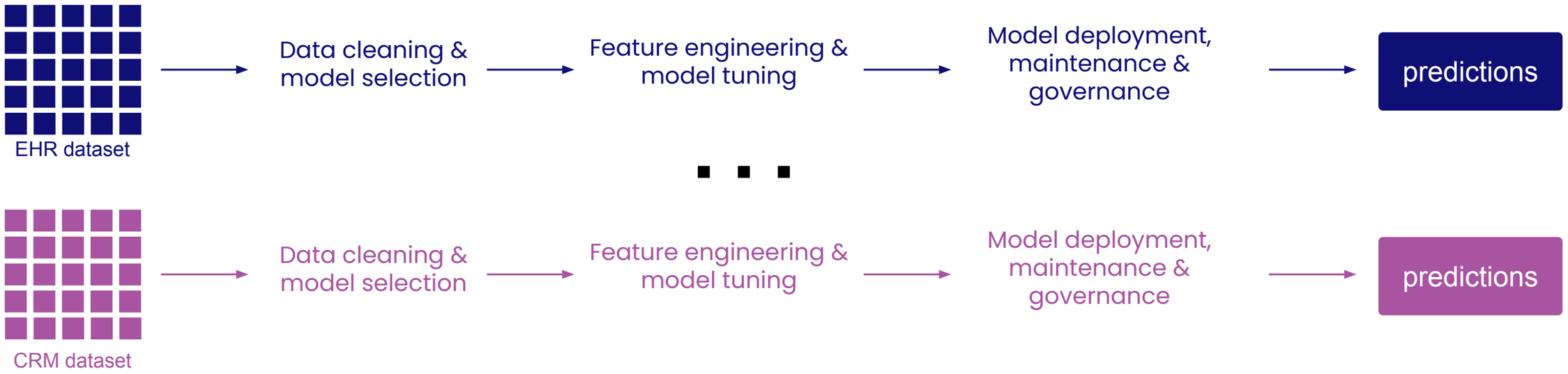
databases CSVs spreadsheets time series

PRIOR

- Excel at *numbers, statistics & patterns*
- Generate *predictions, forecasts, scores*
- **Directly plug into existing ML value chain**

Traditional ML doesn't scale – nothing compounds

Each prediction task requires its own pipeline, infra and team



3–6 months and a dedicated ML team – per problem

Zero transfer: models learn nothing from each other

60% of ML engineering spent on maintenance

Foundation models to the rescue

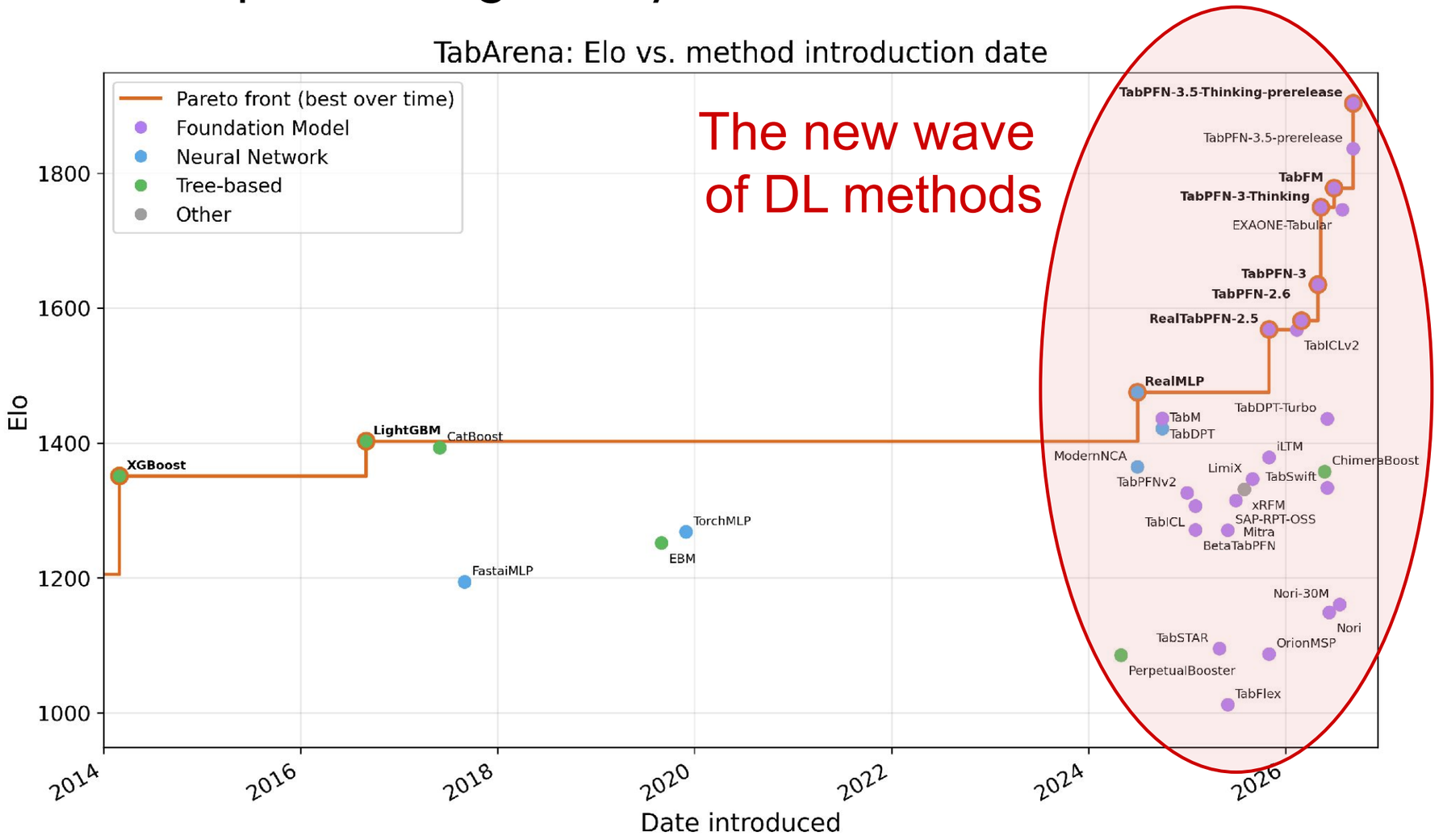
One model can replace all these custom pipelines: instant predictions in a forward pass




```
clf = TabPFNClassifier().fit(X_train, y_train)
predictions = clf.predict(X_test)
```

Helping data scientists become “10x data scientists”

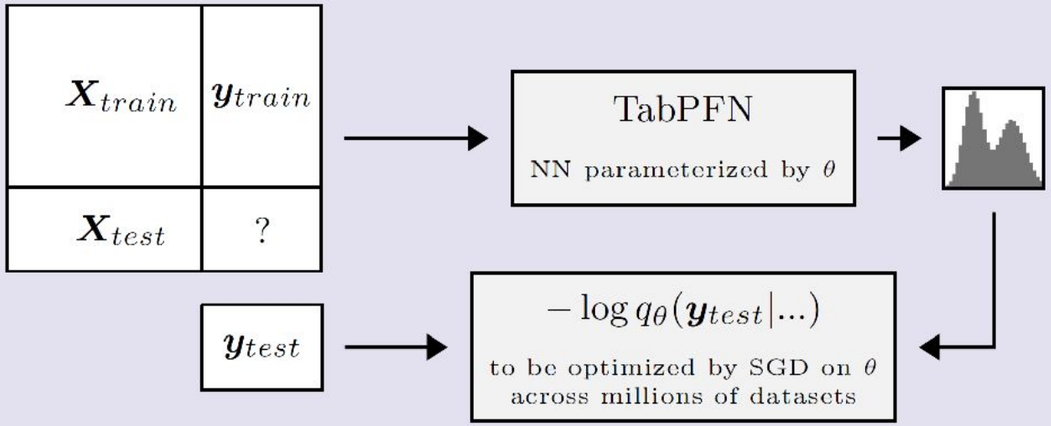
Deep Learning finally excels for tabular data



TabArena classification (benchmark with datasets up to 100k data points); https://tabarena.ai

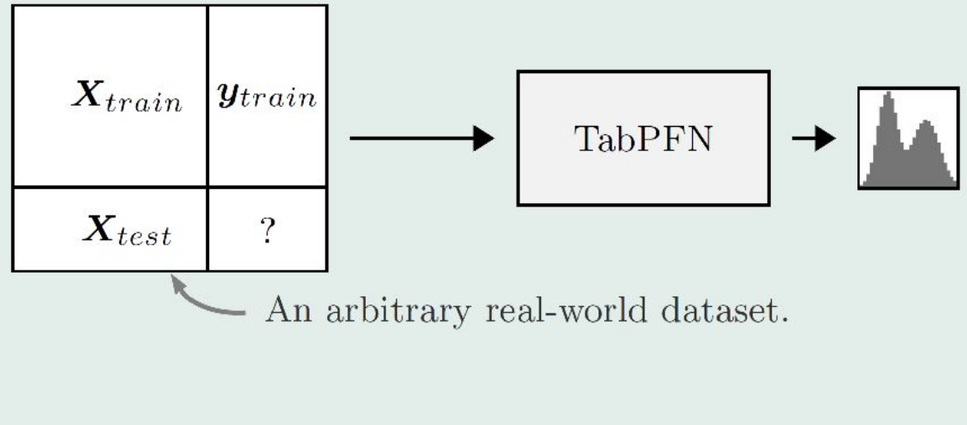
TabPFN overview of pretraining & inference

TabPFN is pretrained on synthetic data to take entire datasets as inputs and predict in a forward pass.



Slow, done by the model provider once per release

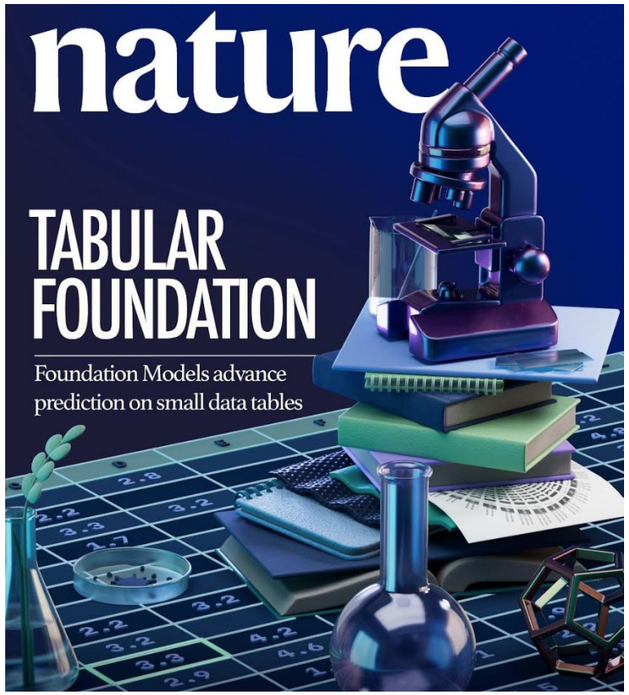
The pretrained TabPFN can be applied out-of-the-box to arbitrary unseen real-world datasets.



Fast; one forward pass on the real data

TabPFN v2: deep learning finally made it for tabular data

Nature's #1 most cited paper published in 2025, along with DeepSeek-R1



Article | Open access | Published: 08 January 2025

Accurate predictions on small data with a tabular foundation model

Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister & Frank Hutter

Nature **637**, 319–326 (2025) | Cite this article

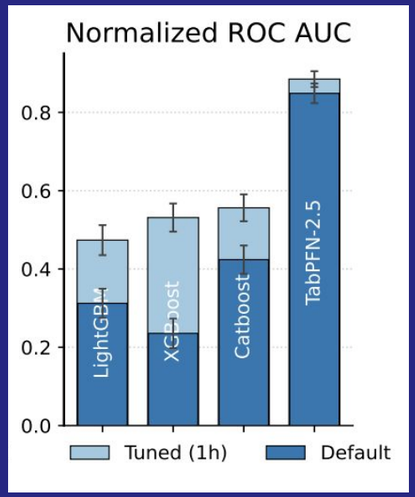
432k Accesses | **438** Citations | **515** Altmetric | Metrics

Alejandro Saucedo | KubeCon 2025 AI Day Keynote
@AxSaucedo

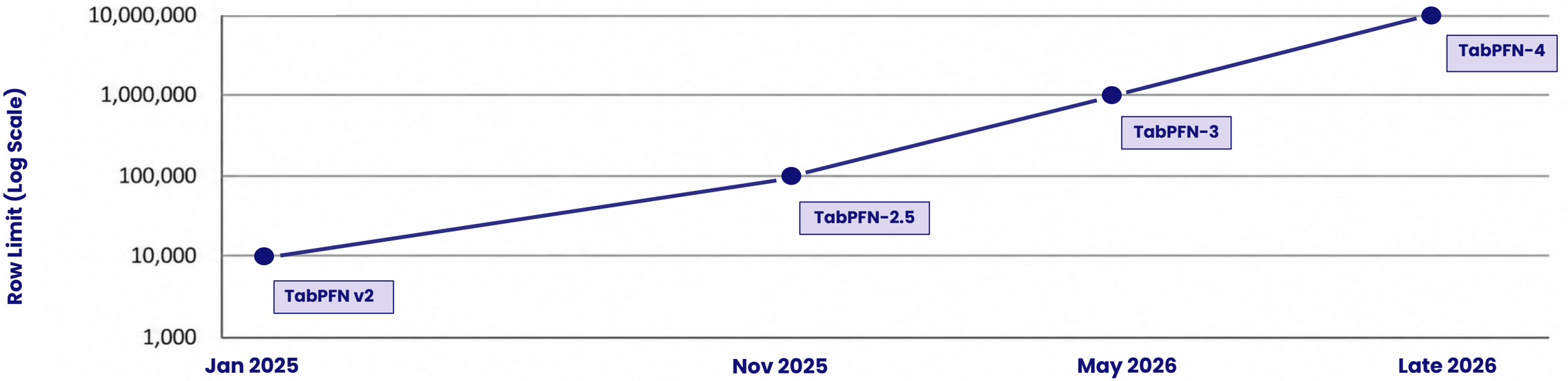
We've entered the ImageNet moment for Tabular Data! A new Tabular Foundational Model is released with TabPFN 2.5 tackling zero-shot predictions; and the sky is the limit in a world that runs on tabular data (spreadsheets anyone?):

GitHub 6K+ STARS

3M+ DOWNLOADS



TabPFN size limit over the years



1000x increase in row limit in less than 2 years

TabPFN is easy to use

It can be accessed in a variety of ways



TabPFN is used in thousands of applications

Prediction tasks in healthcare, finance, industrials, energy, tech and many more

HEALTHCARE Drug response	ENERGY Load forecasting	INDUSTRIALS Predictive maintenance	BUSINESS ANALYTICS Lead scoring	FINANCE Market Forecasting
SUPPLY CHAIN Resource planning	MANUFACTURING Quality control	TabPFN	TRADING Signal generation	PHARMACEUTICAL Clinical trial forecasting
INSURANCE Fraud detection	INDUSTRIALS Site Operations	BUSINESS ANALYTICS Sales forecasting	BANKING Credit risk scoring	TECH Customer lifetime value